

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	06 July 2026
Team members	Jyothi swaroop,bhu Mahesh,mohith
Project Name	MERN Stack Customer Management Application
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

TECHNICAL ARCHITECTURE DIAGRAM MERN Stack Customer App

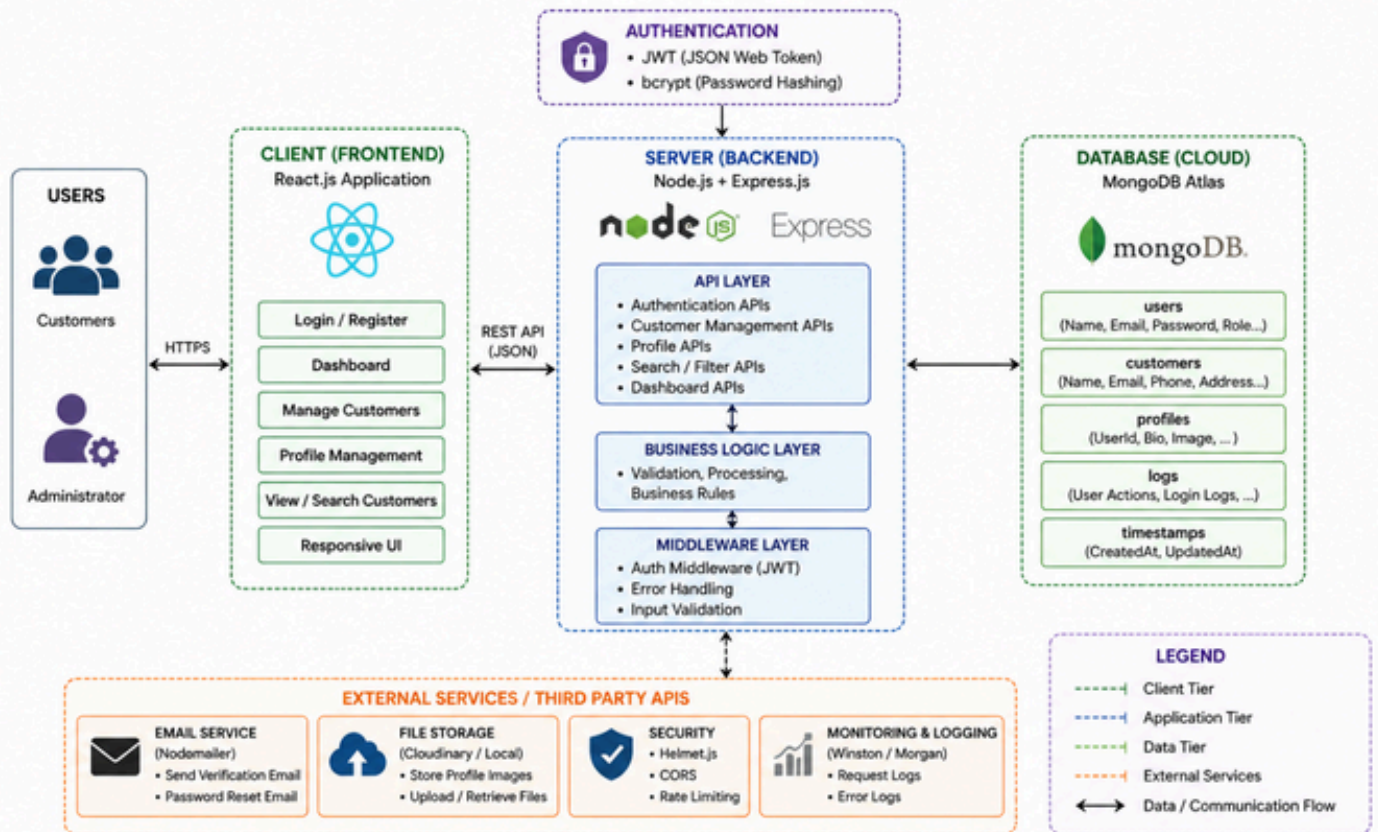


Table 1 – Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Customer interface for login, registration and dashboard	React.js, HTML5, CSS3, Bootstrap
2	Application Logic-1	Authentication and Authorization	Node.js, Express.js, JWT
3	Application Logic-2	Customer Management (CRUD)	Express.js REST APIs
4	Application Logic-3	Profile Management & Validation	Express.js, Mongoose
5	Database	Stores customer information	MongoDB
6	Cloud Database	MongoDB Cloud Database	MongoDB Atlas
7	File Storage	Profile images (optional)	Local Storage / Cloudinary
8	External API-1	Email Verification	Nodemailer
9	External API-2	Password Encryption	bcrypt.js
10	Machine Learning Model	Not Applicable	—
11	Infrastructure	Application Hosting	Render / Vercel / Localhost

Table 2 – Application Characteristics

Characteristic	Description	Technology
Open Source Frameworks	Frontend & Backend Frameworks	React.js, Express.js, Node.js
Security Implementations	Password hashing, JWT Authentication, Input Validation	bcrypt.js, JWT, Helmet, Express Validator
Scalable Architecture	Three-tier Architecture	React + Express + MongoDB
Availability	24×7 cloud deployment	Render, MongoDB Atlas
Performance	Fast API responses, optimized database queries	Express.js, MongoDB Indexing